

Solve the following system:

E-Vex

$$\begin{cases} 3x + 2y = 23 \\ \ln x - \ln y = 2 \ln \sqrt{7} \end{cases}$$

$$\ln x - \ln y \Rightarrow \ln\left(\frac{x}{y}\right) \quad , \quad 2 \ln \sqrt{7} = \ln \sqrt{7}^2$$

$$\text{then: } e^{\ln\left(\frac{x}{y}\right)} = e^{\ln \sqrt{7}^2}$$

$$\frac{x}{y} = 7$$

$$\boxed{x = 7y}$$

$$\rightarrow 3(7y) + 2y = 23$$

$$23y = 23$$

$$\boxed{y = 1}$$

$$x = 7(1)$$

$$\boxed{x = 7}$$

$$S: \{(7, 1)\}$$

E-Vex